

ADITYA PAREEK

Mysuru, Karnataka

☎ +91-8829099340 ✉ adpaniitian@gmail.com 🔗 LinkedIn 🌐 GitHub 📁 Portfolio

Education

The National Institute of Engineering (NIE), Mysuru

Aug 2023 – Jun 2027

Bachelor of Engineering in Information Science & Engineering

- Relevant Coursework: Database Management Systems, Operating Systems, Data Structures & Algorithms

Indian Institute of Technology (IIT), Madras

Jun 2023 – Jun 2027

Bachelor of Science in Data Science

- Relevant Coursework: System Commands, Business Data Management

Technical Skills

Languages: JavaScript, Python, Java, C

Frontend: React, TailwindCSS, EJS, Responsive Design, Component Architecture

Backend: Express.js, FastAPI, Node.js, RESTful APIs, Microservices Architecture

Databases: MySQL, MongoDB, Firestore, SQLite3, Database Design, Query Optimization

AI/ML: OpenCV, Random Forest, LangChain, Google Gemini, Matplotlib, Seaborn

DevOps & Tools: Docker, Firebase, Git, GitHub, Vercel AI SDK, CI/CD, Netlify

Projects

Scalable Video Intelligence System | *React, Express.js, OpenCV, Firebase* | GitHub

Mar 2025 – Present

- Architected full-stack AI surveillance platform processing **15+ concurrent video streams** with real-time analysis, **cutting manual surveillance workload by 70%**
- Built **RESTful microservices architecture** using Express.js with independent scaling for video processing, user management, and analytics modules
- Implemented ML-based facial recognition pipeline with OpenCV achieving **95%+ accuracy** across 200+ users, **eliminating 100% of manual attendance errors** and saving 10+ hours/week
- Engineered **role-based access control (RBAC)** system with Firebase Authentication serving 3 user tiers, designed responsive React dashboard with real-time data synchronization via Firestore

Adi Assistant — AI-Powered Coding Platform | *Node.js, Vercel AI SDK, Docker*

Jan 2025 – Present

- Built production-grade browser-based IDE serving **500+ developers** with zero local setup, featuring terminal output streaming, Git integration, diff viewer, and ZIP export capabilities
- Designed **multi-LLM orchestration system** integrating OpenAI, Gemini, and DeepSeek APIs with intelligent fallback routing using Vercel AI SDK, **reducing API timeout failures by 85%**
- Implemented **dynamic token optimization** with context pruning and streaming response architecture, **reducing API costs by 40%** while maintaining response quality
- Engineered **Docker containerization** pipeline with automated Netlify deployment, **cutting deployment time from 30 minutes to 90 seconds**

Data Analyst Agent | *Python, FastAPI, LangChain, Google Gemini* | GitHub

Nov 2024 – Jan 2025

- Built autonomous data analysis platform with **FastAPI backend** converting natural language queries into executable Python code, **automating 80% of exploratory data analysis workflow**
- Architected **sandboxed Python runtime** with memory and timeout constraints for secure code execution, preventing infinite loops and resource exhaustion
- Integrated LangChain with Google Gemini for **NLP-to-code generation pipeline** supporting SQL queries, web scraping, and visualization generation with **2s response time**
- Implemented conversational dataset exploration interface with real-time Matplotlib/Seaborn visualization generation and context retention across queries

Leadership & Achievements

Chess Club Lead | *Corbett Knight (IIT Madras)*

Ongoing

- Organized 5 tournaments with **100+ participants**, managed 8-person committee, **increased club engagement by 150%**

Hackathon Veteran

Multiple Events

- Built **6+ innovative web and AI-based solutions** across multiple hackathons, consistently delivering functional MVPs in **24 hours** under deadline pressure